Technical Reference: SGR Public Trip Search API Endpoint v1

Authored By: Isaiah Nyalali,
Version: 1.1
Last Updated: September 24th, 2025

Abstract

This document provides a technical deep-dive into the unofficial public-facing API endpoint for querying trip schedules and seat availability on the Tanzania Railway Corporation (TRC) Standard Gauge Railway (SGR). The information herein is derived from empirical analysis and is intended for developers and system architects. As this is not official documentation, all endpoints and data structures are subject to unannounced changes by TRC.

---

1. Endpoint Specification

- **HTTP Method:** POST
- **URL:** https://sgrticket-api.trc.co.tz/TICIDIS/api/v1/Public/SearchTrip
- **Authentication:** Not required.

1.1 HTTP Headers

While many headers may be accepted, the following are the minimal viable set required for a successful 200 OK response.

Header	Value	Description
Content-Type	application/json	**Mandatory.** Specifies that the request body is a JSON object.
Accept	application/json	**Mandatory.** Informs the server that the client expects a JSON response.

Table 1: HTTP Headers

2. Request Body Payload

The body of the POST request must be a JSON object containing the trip search parameters. All fields have been observed to be mandatory.

2.1 Payload Parameters

Field	Data Type	Constraint	Description	Example
boardingStationId	Integer	Required	The unique identifier for the origin station. See Station ID table below.	1
landingStationId	Integer	Required	The unique identifier for the destination station. See Station ID table below.	12
date	String	Required	The date of travel in YYYY-MM-DD format. This must be within a 7-day future window.	"2025-09-25"
isOneWay	Boolean	Required	Must be true for this endpoint's primary function.	true
passengerCount	Integer	Required	Number of travelers. The API appears to validate this against total emptySeats.	1
departureDate	String	Required	The full ISO 8601 UTC timestamp of the travel date. Time part is static.	"2025-09-25T00:00:00.000Z"
returnDate	String	Required	**Endpoint Quirk:** This field is mandatory even if isOneWay is true. The same value as departureDate is a safe and valid entry.	"2025-09-25T00:00:00.000Z"
isReservation	Boolean	Required	Must be false for an immediate availability query.	false
loginType	Integer	Required	Static value of 2, presumed to signify a public, unauthenticated user type.	2

Table 2: Payload Parameters

2.2 cURL Example (Strict)

This example demonstrates the minimal required fields for a successful query from Dar Es Salaam to Dodoma.

```
curl 'https://sgrticket-api.trc.co.tz/TICIDIS/api/v1/Public/SearchTrip' \
  -X POST \
  -H 'Content-Type: application/json' \
  -H 'Accept: application/json' \
  --data-raw '{
    "boardingStationId": 1,
    "landingStationId": 12,
    "date": "2025-09-25",
    "isOneWay": true,
    "isReservation": false,
    "loginType": 2,
    "passengerCount": 1,
    "departureDate": "2025-09-25T00:00:00.000Z",
    "returnDate": "2025-09-25T00:00:00.000Z"
  }'
```

Figure 1: cURL Example

3. Station Reference Table

The following is a non-exhaustive list of known station IDs used by the API.

ID	Station Name
1	Dar Es Salaam
2	Pugu
3	Soga
4	Ruvu
5	Ngerengere
6	Morogoro
7	Mkata
8	Kilosa
9	Kidete
10	Gulwe
11	Igandu
12	Dodoma

Table 3: Station Reference Table

4. Response Data Structure

A successful response is a JSON object containing transaction metadata and the primary data payload in the data key.

4.1 Root Object Structure

Key	Data Type	Description
ok	Boolean	true for success. false if the query failed or returned no results.
code	Integer	HTTP status code mirrored. Typically 200.
message	String	Server status message. GNRL100 - Success. on success.
data	Array	An array of Trip objects. Can be an empty array [] if no trips match the criteria.

Table 4: Root object structure

4.2 The Trip Object (inside data array)

Each object in the data array is a comprehensive model of a single available train journey.

Key	Data Type	Description	Example
tripId	Integer	Primary key for this specific trip definition.	88
tripName	String	Human-readable name of the train/service.	"DAR - DOM / ORDINARY LINE"
date	String	Departure date with timezone offset.	"2025-10-02T00:00:00+03:00"
startTime	String	Departure time in HH:mm:ss format.	"18:55:00"
endTime	String	Arrival time at the landingStationId in HH:mm:ss format.	"23:01:00"
boardingStationId	Integer	ID of the origin station for this leg.	1
landingStationId	Integer	ID of the destination station for this leg.	12
tripsPrice	Float	Fare for one adult in the default/lowest class. May be 0.0 for some trips.	31500.0
tripsPriceCurrency	String	Currency code.	"TZS"
railwayCars	Array	An array of Railway Car objects (see 4.3). This is the availability data.	[...]
routeId	Integer	Identifier for the train's overall route.	59
trainSetId	Integer	Identifier for the physical set of wagons being used.	20

Table 5: The trip object

4.3 The Railway Car Object (inside railwayCars array)

This object is the most granular level of detail, representing a single coach on the train.

Key	Data Type	Description	Example
railwayCarId	Integer	Unique ID for the physical car model.	19
typeId	Integer	ID representing the class type.	151 (Royal Class)
typeName	String	**Crucial Field.** The human-readable class name.	"Royal Class"
capacity	Integer	Total seat capacity of this car.	44
emptySeats	Integer	**Crucial Field.** Available seats of all types in this car. 0 means full.	24
hasSeat	Boolean	A flag indicating if the car type has assignable seats.	true
standardSeatCapacity	Integer	The car's total standard seat count.	44
disabledSeatCapacity	Integer	The car's total accessible/disabled seat count.	0
emptyStandardSeats	Integer	Number of available standard seats.	24
emptyDisabledSeats	Integer	Number of available accessible/disabled seats.	0

Table 6: The Railway Car Object

5. Implementation Considerations & Known Behavior

- **API Polling and Rate Limits:** Aggressive, repeated polling from a single IP address may lead to temporary blocking. A responsible client should implement a request backoff strategy or limit polling frequency. Production systems should run these checks from a backend server.
- **Data Aggregation:** A single train trip will often return multiple railwayCars of the same typeName. A client must iterate through the entire railwayCars array and aggregate the emptySeats and capacity values for each typeName (e.g., sum up all "Economy" cars) to present a meaningful total to the user.
- **Endpoint Volatility:** As this is not an officially documented public API, all structures, field names, and even the endpoint URL are subject to change without notice, which could break an integration. Robust error handling and logging are mandatory for any application built on this endpoint.